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## Global Valuation, Accounting &amp; Tax | North America

## AI: Funding the Buildout Beyond the Balance Sheet

Leases, debt, equity, and purchase commitments are extending the data center buildout beyond self-funded hyperscaler cash flow. Each new arrangement, whether on-balance sheet or not, requires committing future cash, and further capex revisions require incremental sources of capital.

## Key Takeaways

- Hyperscalers are expected to spend more than \$1.2tn on cash capex in 2027, versus an estimated \$1tn of total operating cash flow.
- Hyperscalers are using leases, issuing public debt, halting share repurchases, and issuing equity to fund incremental capex.
- More than \$3.1tn of commitments and guarantees from hyperscalers + NVDA and AVGO are enabling third parties to obtain financing to fund AI infrastructure.
- Capex is understated, and FCF is overstated to the extent that operating leases are used to obtain data center construction financing via SPVs.
- If financing costs increase with incremental commitments, returns on new AI infrastructure investment will need to cover the rising cost of capital.

The rush to build compute infrastructure to meet rising demand has resulted in investment levels well in excess of cash generated in operations, which are largely being funded by private credit loans to special purpose vehicles (SPVs) and, more recently, hyperscaler debt and equity issuances.

**The need for capital to support the build:** In order to achieve the lowest financing costs possible, hyperscalers are providing various forms of credit support for the data center buildout, many of which are off-balance sheet today (see [Off-Balance Sheet, On the Hook](#)).

Hyperscalers are also committing to larger and longer-dated purchase commitments to secure GPUs, memory, and networking equipment to be available when data centers are completed. As of their latest SEC filings, hyperscalers' undiscounted commitments total more than \$2.7tn, or ~3 years of current operating cash flow.

Now, chip companies are also stepping in to [lend their balance sheets](#) in further support of financing compute capex as seen in [Broadcom's announced](#) chip leasing facility to support up to 20GW of compute capex and reports of new structures from Nvidia. As investors look to how high capex could go, discussions have shifted to what the source and cost of the next leg of funding will be.

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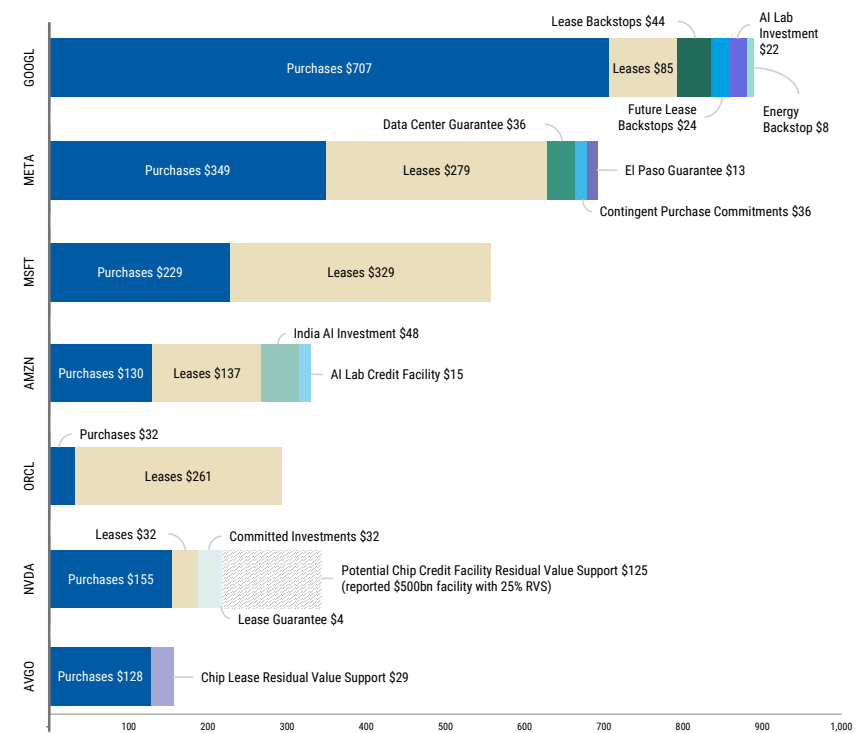
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**Exhibit 1:** Hyperscalers, NVDA, AVGO disclosed off-balance sheet commitments total more than \$3.1tn, with more financing structures under development

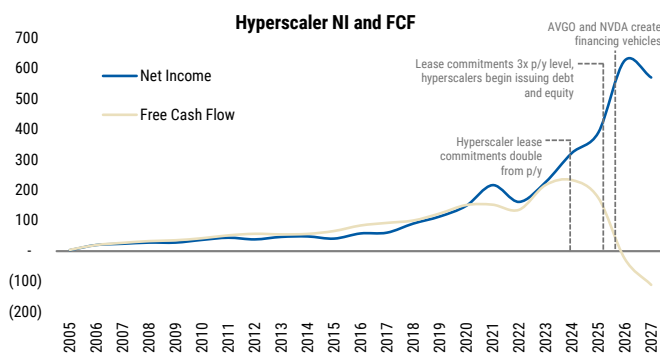


Source: Company Filings, Morgan Stanley Research.

# Funding the Build Beyond Operating Cash Flow

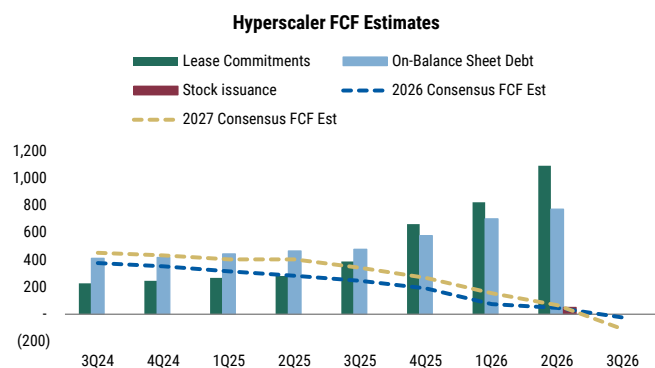
**Hyperscalers are reinvesting more than 40% of their sales back into AI capex** (see [Capital Intensity Enters a New Era](#)). Despite operating high-margin legacy businesses, this level of investment cannot be sustained solely from current operating cash flow. Amazon and Google FCF turned negative in 2Q26, and Meta is expected to turn negative next quarter. As FCF turns negative, hyperscalers are turning to leasing, issuing debt, and halting buybacks and issuing shares to fund their compute capacity goals.

**Exhibit 2:** Hyperscaler cash flow no longer covers the compute build...



Source: Factset, Morgan Stanley Research. Consensus estimates as of August 14, 2026.

**Exhibit 3:** ...new sources of funding first from lease commitments, then debt, and recently equity.



Source: Factset, Company Filings, Morgan Stanley Research.

## Initial External Funding Source: Hyperscaler Commitments Support Financing for Suppliers

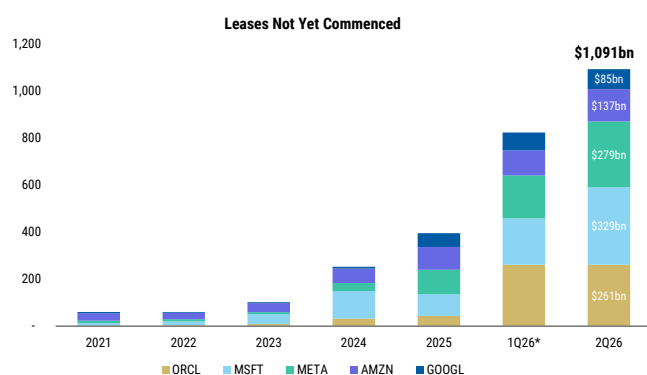
Suppliers and data center developers can borrow against long-dated leases, guarantees, or purchase commitments from investment grade hyperscalers, allowing capacity to be built before the hyperscalers make any payments or recognize liabilities. These structures are [off-balance sheet](#) and, as a result, cash is committed sooner than on-balance sheet leverage would suggest.

Hyperscalers' first source of external funds was leases. A substantial portion of data center shells are being funded via Special Purpose Vehicles (SPVs) that raise capital from private credit, supported by a lease from an investment-grade hyperscaler. As of their latest filings, hyperscalers have committed to \$1.1tn of lease payments for leases that have not yet started ([Exhibit 4](#)). See accounting for [Off-Balance-Sheet Project Debt](#) below.

Hyperscalers are also entering into large, long-dated purchase commitments for GPUs, memory, manufacturing capacity, networking and other equipment for these data centers upon construction completion. Nvidia and Broadcom have similarly begun securing fab capacity for their inventory. These purchase commitments now total more than \$1.7tn ([Exhibit 5](#)).

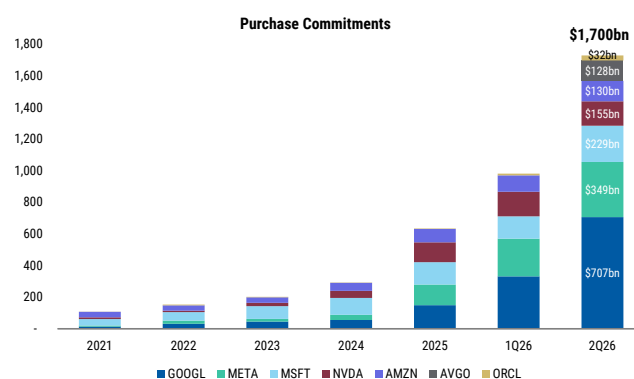
The commitments under each of these structures will show up in free cash flow once completed, but the increasing size and duration adds to operating leverage. These contracts are strategically important while supply remains constrained, but if supply and demand normalize ahead of expectations, companies may have to pay for excess capacity or attempt to renegotiate the terms.

**Exhibit 4:** Hyperscalers have committed to \$1.1tn of undiscounted lease payments.



Source: Company filings, Morgan Stanley Research.

**Exhibit 5:** NVDA, AVGO and the hyperscalers disclose \$1.7tn of purchase commitments.



Source: Company filings, Morgan Stanley Research.

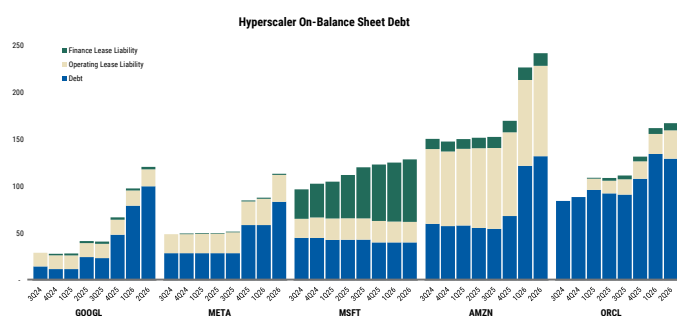
## Next Layer of Financing: Bond Issuance and On-Balance Sheet Debt

On-balance sheet debt has also been rising at hyperscalers. As their data centers are completed, the related operating or finance lease debt is brought on balance sheet. In addition to leases, hyperscalers began issuing debt in late 2025 and accelerated issuances throughout this year. On-balance sheet debt and lease liabilities now totals \$770bn across the hyperscalers, inclusive of long- and short-term debt and lease liabilities ([Exhibit 6](#)).

**Hyperscaler issuance has risen from 2% of nonfinancial investment-grade supply in 2025 to 19% in 2026 YTD** ([Exhibit 7](#)). As capex revisions are increasingly funded by debt instead of cash, the economics of these projects become increasingly dependent on the cost of external capital. Our credit strategy team has been tracking the widening spreads across AI financing [here](#).

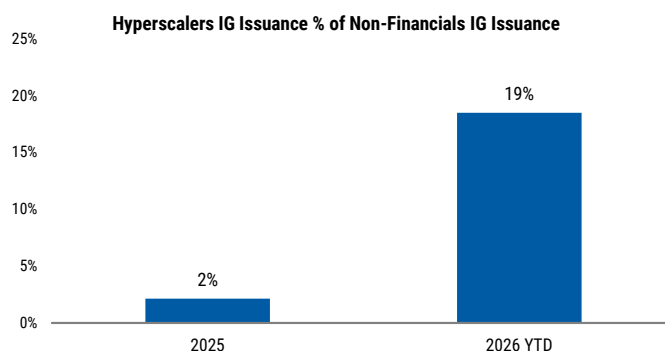
While on-balance sheet liabilities have been gradually increasing, they still pale in comparison to off-balance sheet commitments ([Exhibit 8](#)). Hyperscalers are committing to buy compute equipment and to support future revenue ([Exhibit 9](#)). Meta and Google are building compute to support internal models, so commitments exceed RPO, but for compute sellers MSFT and AMZN, commitments are approaching RPO.

**Exhibit 6:** Hyperscaler on-balance sheet debt has risen to \$770bn.



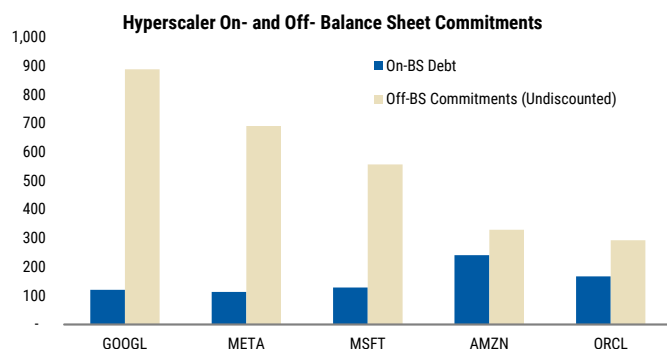
Source: Dealogic, Company Filings, Morgan Stanley Research.

**Exhibit 7:** Hyperscaler issuance accounts for 19% of total non-financial IG issuance.



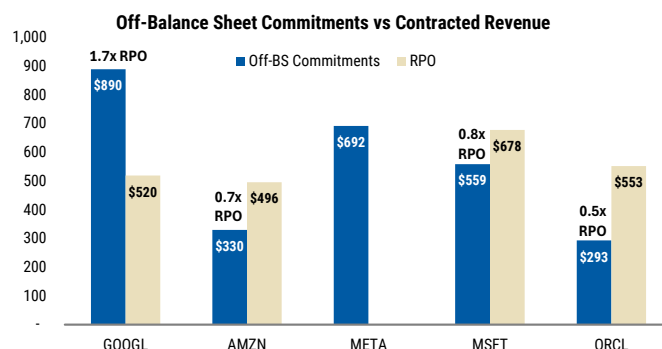
Source: Dealogic, Morgan Stanley Research.

**Exhibit 8:** Reported leverage remains modest relative to the amount of off-BS commitments.



Source: Company Filings, Morgan Stanley Research.

**Exhibit 9:** Off-balance sheet commitments are approaching contracted revenue for compute providers.



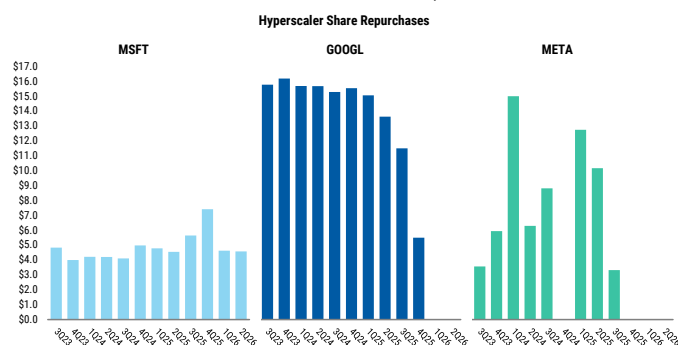
Source: Company Filings, Morgan Stanley Research.

## Third Funding Lever: Lower Buybacks and Equity Issuance

Hyperscalers have relied more on leases and debt than on equity to fund their capex ambitions, but equity could become more useful if interest rates rise, credit spreads widen, or AI investment remains elevated for longer than expected. Before issuing shares, hyperscalers are preserving cash by reducing buybacks, allowing for stock-based compensation dilution. Microsoft is the only hyperscaler that has yet to cut its share buybacks or issue debt ([Exhibit 10](#)).

**Underappreciated dilution:** A significant portion of historical share repurchases was intended to offset equity compensation dilution, but that offset is waning. For instance, Google has cut its share buybacks from more than \$60bn a year to zero and issued \$50bn of equity in 2Q26, resulting in more than \$110bn more cash to invest in AI infrastructure. Shareholders are now beginning to experience dilution after the company had reduced its share count by 13% from its peak in the last decade.

**Exhibit 10:** MSFT continues to offset dilution through buybacks, while GOOGL and META have reduced repurchases...



## Emerging Funding Sources: Customer Prepayments & Chip Vendor Support

### Customer Prepayments: Oracle

Oracle has started to use customer prepayments as a source of capex financing. Oracle's data centers are built by SPVs created to develop the site; chips have been purchased with cash, either from operating cash flows, debt issuance, or equity issuance. In the most recent quarter, Oracle disclosed \$4.6bn of customer prepayments to fund capex.

These prepayments are considered deferred revenue on the balance sheet, though more debt-like than typical deferred revenue that does not have a financing component (meaning the prepayment is received more than a year in advance of when the revenue will be recognized). Interest expense will be charged to the P&L based on Oracle's incremental borrowing rate. Oracle's 10-year and 30-year bonds are currently trading at 6.9% and 7.9% implied ytw, and interest expense on customer prepayments should reflect a similar borrowing rate.

**Investor Insight:** Interest accretes on the customer prepayment in deferred revenue until the revenue is recognized. For example, if a customer makes a \$10bn 2-year advance payment for a future purchase of goods, and the seller has a 10% incremental borrowing rate, the seller would recognize ~\$1bn of interest expense each year (compounded in year 2), and ~\$12bn of revenue upon delivery of the goods to the customer.

### Chip Leasing SPVs Backed by Chip Vendor Residual Value Support: Broadcom and Nvidia

Broadcom and Nvidia have announced SPV chip financing structures to help their customers secure financing to access chips. Up until recently, chips were mainly purchased with hyperscaler cash flows. Now, new chip SPVs are issuing debt to purchase chips and lease them to unrated AI labs. Broadcom and Nvidia are providing residual value support to cover the SPV for any shortfalls if lease payments lapse and resale is not sufficient to repay specified bond holders. These financing vehicles supported by chip vendors allow unrated AI labs to lease chips from SPVs at a funding cost closer to that of their IG suppliers. See our credit team's analysis of [balance sheet-as-a-service](#) for more detail.

**Investor Insight:** Depending on the structure of these arrangements, chip vendors may need to split the proceeds from sales to these SPVs between (1) the chips they are selling and (2) the residual value guarantee they are providing. When a seller provides residual value support, the seller/guarantor should recognize a guarantee liability at the time of sale. The guarantee liability is measured at "fair value" as the premium that would have been received if the same guarantee were issued in a standalone transaction. In the event that the guarantee never requires payment, guarantee income would generally not be considered revenue.

# Off-Balance-Sheet Project Debt

## Accounting for Data Center Financing SPVs

Data centers constructed for hyperscalers to lease are off-balance sheet during construction. Hyperscalers, in partnership with private credit, create Special Purpose Vehicles (SPVs) to finance data center developments. These SPVs are considered "variable interest entities" (VIE) for accounting purposes because as the sole tenant of the data center, hyperscalers have more exposure to and influence on the SPV than their minority equity ownership and voting rights would suggest.

VIE structures became increasingly common and used to circumvent consolidation in the late '90s and early '00s. In response, consolidation accounting was updated so that VIEs would be consolidated by the SPV's "primary beneficiary". For a hyperscaler to be considered the primary beneficiary of a lease SPV, it must have both (1) *the power to direct the most economically significant activities* and (2) *the obligation to absorb the entity's losses or the right to receive the entity's economic benefits*.

The following table highlights considerations for whether or not a hyperscaler needs to consolidate a lease SPV:

| Power                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Losses and Benefits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Leasing decisions:</b> Who decides which hyperscaler, neocloud, or AI lab will occupy the facility and on what terms? Leasing decisions are weighed more heavily for shorter-duration leases where a larger portion of the economic life needs to be re-leased at the end of the initial term.</p> <p><b>Operating decisions:</b> Who controls the day-to-day operations, contracts with suppliers, and oversees maintenance and upgrades that impact the fair value of the data center?</p> <p><b>Marketing decisions:</b> Who decides how to remarket the data center at the end of the hyperscaler's lease?</p> | <p><b>Residual value risk:</b> When hyperscalers guarantee the value of a data center at the end of a lease term or a minimum lease rate for the remainder of data center life, they are considered to have risk of loss.</p> <p><b>Obligations to fund construction cost overruns:</b> If the hyperscaler bears losses from significant cost overruns, there is exposure to the risk of the SPV's losses.</p> <p><b>Credit risk:</b> Tenant credit risk can be considered significant in determining whether or not to consolidate as lease SPV, but the risk would not be significant for a high quality hyperscaler tenant.</p> |

Meta and Google both disclose that they do not consolidate data center SPVs because they determined they do not have power to direct the VIE's activities (see table below for disclosures). For example, something as simple as Meta's lack of control over negotiations with future lease tenants and property sales is sufficient to not consolidate the SPV under the primary beneficiary model, and related debt remains off balance sheet until the lease begins.

**Investor Insight:** *What's off-balance sheet today, may not be tomorrow.* The VIE model requires significant judgment, and the initial determination is not static over the life of the transaction. GAAP requires continual reassessment of the power and economics tests. For example, the consolidation conclusion could change as the likelihood of needing to provide residual value support changes. This accounting judgment affects the reported balance sheet, leverage, and how the arrangement appears to investors.

| META VIE Footnote (2Q26 10-Q, emphasis added)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | GOOGL VIE Footnote (2025 10-K, emphasis added)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Significant judgment is required to identify the activities that most significantly impact the Venture's economic performance based upon the purpose and design of the entity. This judgment included, but was not limited to, considering future conditions that may impact the fair value of the Venture's long-lived assets (including expectations of payments under the RVG) or the Venture's ability to generate cash flows. On the basis of analyses performed, <b>decisions pertaining to remarketing the data center campus</b>, including but not limited to, <b>negotiations with future lease tenants and individual property sales</b>, were determined to have the most significant impact on the Venture's economic performance. <b>As we do not have the power to direct the activities</b> that most significantly impact the Venture's economic performance, <b>we are not the primary beneficiary and, therefore, do not consolidate the variable interest entity (VIE)</b>. Our ongoing involvement with the VIE includes providing construction management, administrative and property management services to the Venture.</p> | <p>We hold various forms of interests in VIEs, including certain of our investments in private companies and renewable energy entities, certain leases and credit backstops with data center entities, and certain backstops with energy infrastructure entities. Because we have determined that <b>we do not direct the activities that most significantly impact the economic performance of these entities</b>, we are not the primary beneficiary. <b>Therefore, these VIEs are not consolidated within our financial statements.</b></p> |

# Microsoft Lease Accounting Change in Estimate

Last quarter, MSFT announced they are extending the useful lives of their data centers from 15 to 25 years. As a result, their data center leases will shift from finance leases to operating leases. Leases are classified as finance leases if they meet certain criteria (see our full lease accounting note [here](#)). One of the tests is whether the lease covers 75% or more of the data center's life. New construction data centers are typically leased for 15 years (or more), which was 100% of MSFT's old useful life estimate and only 60% of the new useful life estimate.

MSFT presents FCF inclusive of finance lease capex but excludes operating leases from the measure. Therefore, capex will look lower, and FCF will look higher as a result of the new accounting estimate. Economically, the leases have not changed substance and still function similarly to funding capex with debt.

Additionally, we would consider the following two common financing transactions to be economically similar:

- One company creates an SPV to build a data center and agrees to lease the data center for 15-17 years, with lease payments sufficient to pay back construction debt over that horizon (or other credit enhancement features like a residual value guarantee that would provide a specified return to bondholders).
- Another company builds the data center itself, using cash from bond issuance to fund the construction.

Both transactions require external financing and cash flows that provide an acceptable return to lenders. In order to compare FCF between companies engaging in these transactions, investors can adjust capex and FCF to include the value of the data centers acquired under long-term operating leases and add back operating lease cost to treat it similarly to depreciation, both of which are available in the lease footnote. Note that regardless of the lease classification, credit rating agencies treat lease liabilities as debt.

**Exhibit 12:** MSFT's lease footnote in the FY2026 10-K shows \$24.6bn of assets acquired under *finance* leases last year...

**NOTE 13 — LEASES**

We have operating and finance leases for datacenters, corporate offices, research and development facilities, and certain equipment. Our leases have remaining lease terms of less than 1 year to 20 years, some of which include options to extend the leases for up to 5 years, and some of which include options to terminate the leases within 1 year.

The components of lease expense were as follows:

| (In millions)                       | 2025     | 2025     | 2024     |
|-------------------------------------|----------|----------|----------|
| Year Ended June 30,                 |          |          |          |
| Operating lease cost                | \$ 6,968 | \$ 5,524 | \$ 3,555 |
| Finance lease cost:                 |          |          |          |
| Amortization of right-of-use assets | \$ 5,493 | \$ 3,498 | \$ 1,900 |
| Interest on lease liabilities       | 2,547    | 1,417    | 734      |
| Total finance lease cost            | \$ 7,960 | \$ 4,825 | \$ 2,534 |

Supplemental cash flow information related to leases was as follows:

| (In millions)                                                           | 2025     | 2025     | 2024     |
|-------------------------------------------------------------------------|----------|----------|----------|
| Year Ended June 30,                                                     |          |          |          |
| Cash paid for amounts included in the measurement of lease liabilities: |          |          |          |
| Operating cash flows from operating leases                              | \$ 6,443 | \$ 4,931 | \$ 3,550 |
| Operating cash flows from finance leases                                | 2,547    | 1,372    | 734      |
| Financing cash flows from finance leases                                | 3,101    | 2,283    | 1,286    |
| Right-of-use assets obtained in exchange for lease obligations:         |          |          |          |
| Operating leases                                                        | 4,665    | 7,826    | 6,703    |
| Finance leases                                                          | 24,698   | 20,511   | 11,633   |

Source: Company Filings, Morgan Stanley Research.

**Exhibit 13:** ...while ORCL's FY2026 10-K shows \$18.2bn of assets acquired under *operating* leases.

**Leases**

We have operating and finance leases that primarily relate to our data centers and real estate facilities. As of May 31, 2026, our leases substantially have remaining terms of one year to seventeen years, some of which include options to extend and/or terminate the leases.

The components of lease expense were as follows:

| (In millions)                 | 2026     | 2025     | 2024     |
|-------------------------------|----------|----------|----------|
| Year Ended May 31,            |          |          |          |
| Operating lease cost          | \$ 2,794 | \$ 1,716 | \$ 1,159 |
| Finance lease cost:           |          |          |          |
| Amortization of ROU assets    | \$ 351   | \$ 48    | \$ —     |
| Interest on lease liabilities | 279      | 38       | —        |
| Total finance lease cost      | \$ 630   | \$ 86    | \$ —     |

Supplemental cash flow information related to leases was as follows:

| (In millions)                                                           | 2026      | 2025     | 2024     |
|-------------------------------------------------------------------------|-----------|----------|----------|
| Year Ended May 31,                                                      |           |          |          |
| Cash paid for amounts included in the measurement of lease liabilities: |           |          |          |
| Operating leases                                                        | \$ 2,548  | \$ 1,685 | \$ 1,168 |
| Finance leases                                                          | \$ 452    | \$ 27    | \$ —     |
| ROU assets obtained in exchange for lease obligations:                  |           |          |          |
| Operating leases                                                        | \$ 18,246 | \$ 6,970 | \$ 4,246 |
| Finance leases                                                          | \$ 4,946  | \$ 2,921 | \$ —     |

Source: Company Filings, Morgan Stanley Research.

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